Soumitri Chattopadhyay

UC San Diego, CA, USA

https://soumitri2001.github.io | sochattopadhyay@ucsd.edu

Google Scholar | GitHub | LinkedIn

Research Interests

Methods: Foundational Models, Multi-modal learning, In-Context learning, Quantitative 3D/4D Extraction
Applications: Medical Image Computing, Longitudinal Analysis, Clinical AI Workflows, World Models

Education

University of California, San Diego — California, USA
Ph.D. in Computer Science and Engineering — *Sept. 2025 – May 2028 (Expected)*

University of North Carolina at Chapel Hill — North Carolina, USA
M.S. in Computer Science (Ph.D. Transferred to UCSD) — *Aug. 2023 – May 2025*

Jadavpur University — Kolkata, India
B.E. in Information Technology | CGPA (out of 10): 9.01 — *Aug. 2019 – June 2023*

Research Experiences

UC San Diego — California, USA
Graduate Student Researcher — *Sept. 2025 – Present*

- Part of the UCSD Biomedical Image Analysis Group, advised by Prof. Marc Niethammer.
- Working on quantitative extraction for 4D brain perfusion from intra-operative 2D angiography sequences.
- Developing a physics-driven hemodynamic simulator for brain CT perfusion to study ischemic stroke.
- Proposed methods for training-free multi-modal in-context medical segmentation. **(WACV'27, MICCAI'26)**.
- Developed an end-to-end framework to process, segment and extract biomarkers from 2D and 3D raw clinical imaging data for early diagnosis of Lupus (as a part of an **NIH grant**).

UNC Chapel Hill — North Carolina, USA
Graduate Research Assistant — *Aug. 2023 – Sept. 2025*

- Part of the erstwhile UNC-BIAG, advised by Prof. Marc Niethammer. Transferred to UCSD upon advisor's move.
- Conducted an extensive study on zero-shot domain generalizability of medical segmentation foundational models.
- Proposed a few-shot latent diffusion model-aided fine-grained image recognition **(BMVC '24 Oral)**.

SRI International — Menlo Park, CA
Research Intern — *May 2024 – Dec. 2024*

- Interned in the NuSCI Research Group, supervised by Dr. Anirban Roy
- Worked on **brain decoding**; developed an end-to-end framework for visual decoding from fMRI scans with controllable latent diffusion models and various training recipes that outperformed prior SOTA works.

École de technologie supérieure (ÉTS), Montreal — Montreal, Canada
MITACS Globalink Research Intern — *June 2022 – Aug. 2022*

- Advisor: Prof. Jose Dolz
- Explored handling of 3D medical volumes (CT and MRI), including pre-processing, sampling, and visualization.
- Implemented a knowledge distillation-aided baseline for unpaired multi-modal medical image segmentation.

CVPR Unit, Indian Statistical Institute — Kolkata, India
Research Intern — *Nov. 2021 – May 2023*

- Supervisors: Prof. Umapada Pal and Prof. Saumik Bhattacharya
- Investigated the role of temperature in Self-supervised Contrastive loss and proposed a dynamic temperature scaling to improve its performance **(IEEE TAI '24)**.
- Developed Self-Supervised Learning frameworks for Offline Signature Verification **(ICIP '22 Oral, ICPR '22)**.

Preprints/Submitted

5. **S. Chattopadhyay**, B. Demir, Y. Jin, Y. Z. Lee, M. Niethammer, You see Voxels, I see Features: A Unified Pipeline for Reliable Processing of Heterogeneous Clinical Imaging Data, *In Preparation.*
4. **S. Chattopadhyay,** Y. Wang, D. Mandal, B. Demir, M. Niethammer, P. Chakravarthula, DRONE: Diffractive Optical Neural Networks for Ultra–Light-Weight Medical Image Segmentation, *Under Review at ISBI 2027.*
3. Y. Jin, **S. Chattopadhyay**, B. Demir, Y. Z. Lee, M. Niethammer, Bone Segmentation-Guided High-Resolution Adult Hand Skeletal X-ray Synthesis, *Under Review at ISBI 2027.*
2. **S. Chattopadhyay**, B. Demir, Y. Jin, M. Niethammer, MedSegQC: A Data-driven Approach for Quality Assessment of 3D Medical Image Segmentation, *In Preparation.*
1. D. Mandal, **S. Chattopadhyay**, Y. Wang, M. Niethammer, P. Chakravarthula, UnSCAR: Universal, Scalable, Controllable, and Adaptable Image Restoration, *Under Review.*

Selected Publications

15. **S. Chattopadhyay**, Y. Jin, B. Demir, M. Niethammer, UniMedMICS: Generalizing Training-free Universal Multimodal In-Context 3D Medical Image Segmentation, **WACV 2027** *(accepted)*
14. **S. Chattopadhyay**, B. Demir, M. Niethammer, Uncertainty-Aware Spatio-Semantic Contextual Prompts for Multimodal Medical Segmentation, **MICCAI 2026** *(Early Accepted, top–9%)*
13. B. Demir, **S. Chattopadhyay**, H. Greer, B. Chen, M. Niethammer, How Useful Are Vision Foundation Model Features for Out-of-the-Box Disease Progression Prediction?, **CVPR 2026 – CV4Clinic Workshop, Best Paper Award**
12. A. Das, S. Biswas, **S. Chattopadhyay**, A. Lodh, et al., NotesBank: Benchmarking Neural Transcription and Search for Scientific Notes Understanding , **ECML-PKDD 2026**
11. **S. Chattopadhyay**, B. Demir, M. Niethammer, On the Robustness of Foundational 3D Medical Image Segmentation Models Against Imprecise Visual Prompts, **ISBI 2026, Oral**
10. D. Mandal, **S. Chattopadhyay**, G. Tong, P. Chakravarthula, UniCoRN: Latent Diffusion-based Unified Controllable Image Restoration Network across Multiple Degradations, **WACV 2026**, **Oral**
9. **S. Chattopadhyay**, S. Biswas, E. Vivoli, J. Llados, Towards Generative Class Prompt Learning for Fine-grained Visual Recognition, **BMVC 2024**, **Oral**
8. A. Sain, A.K. Bhunia, S. Koley, P.N. Chowdhury, **S. Chattopadhyay**, T. Xiang, Y.Z. Song, Exploiting Unlabelled Photos for Stronger Fine-Grained SBIR, **CVPR 2023**
7. H. Thakur and **S. Chattopadhyay**, Active Learning for Fine-Grained Sketch-Based Image Retrieval, **BMVC 2023**
6. M. Bhattacharyya, **S. Chattopadhyay**, S. Nag, DeCAtt: Efficient Vision Transformers with Decorrelated Attention Heads, **CVPR 2023 – ECV Workshop**, **Oral**
5. **S. Chattopadhyay**, S. Ganguly, S. Chaudhury, S. Nag, S. Chattopadhyay, Exploring Self-Supervised Representation Learning For Low-Resource Medical Image Analysis, **ICIP 2023**
4. A. Deb, S. Nag, A. Mahapatra, **S. Chattopadhyay**, A. Marik, *et al.* BeAts: Bengali Speech Acts Recognition using Multimodal Attention Fusion, **INTERSPEECH 2023**, **Oral**
3. H. Basak, **S. Chattopadhyay**, R. Kundu, S. Nag, R. Mallipeddi, IDEAL: Improved DEnse locAL Contrastive Learning for Semi-Supervised Medical Image Segmentation, **ICASSP 2023**

2. S. Manna, **S. Chattopadhyay**, S. Bhattacharya, U. Pal, SWIS: Self-Supervised Representation Learning For Writer Independent Offline Signature Verification, **ICIP 2022**, **Oral**

1. **S. Chattopadhyay**, S. Manna, S. Bhattacharya, U. Pal, SURDS: Self-Supervised Attention-guided Reconstruction and Dual Triplet Loss for Offline Signature Verification, **ICPR 2022**

ACHIEVEMENTS

- Winner of **"Best Paper Award"** at CV4Clinic Workshop, CVPR 2026. View announcement
- Selected for **Research Week with Google** (Track: Computer Vision) organised by Google Research India in January 2023.
- Recipient of the **MITACS Globalink Research Internship** program for the summer of 2022.
- Winner of **"Best Paper Award"** (Track: Deep Learning) at the 4th International Conference on Machine Intelligence and Signal Processing (MISP) 2022. View certificate

Technical Skills

Frameworks/Libraries: PyTorch, MONAI, PyRadiomics, NumPy, Pandas, Matplotlib
Softwares/Developer Tools: LaTeX, VSCode, Slicer, ITK-SNAP, Android Studio

Academic Services

- **Conference reviewing (regular, since 2022):** CVPR, MICCAI, WACV, ISBI, ICCV, ACM MM, ICPR, BMVC, ACCV, NeurIPS
- **Journal reviewing (since 2021):** Medical Image Analysis - Elsevier **;** Engineering Applications of Artificial Intelligence - Elsevier **;** Computers in Biology and Medicine - Elsevier **;** IEEE Access